Arpita Chaudhari

Industrial Software Engineer · IIoT & OT Systems

PERSONAL DETAILS

Location	Ahmedabad, India
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Languages	English (Professional)

PROFILE

Industrial Software Engineer with hands-on expertise in IIoT systems, WebSCADA platforms, and industrial protocol communication. Builds custom Python-based OPC servers, 21 CFR Part 11-compliant WebSCADA applications, OPC-UA tag management tools, and Windows service-packaged deployments for pharmaceutical and manufacturing environments. Proficient in Angular and FastAPI for full-stack industrial applications; experienced in leading junior engineers and delivering production-grade systems end-to-end. MSc Computer Science, University of Greenwich, London.

PROFESSIONAL EXPERIENCE

04/2026 – present

System Software Engineer — Industrial Automation

Kevision Systems · Ahmedabad, India

- Design and develop industrial automation software — WebSCADA platforms, OPC servers, SCADA interfaces and pharma machine data-logging tools for manufacturing environments.
- Build Python-based backend services (FastAPI, Django) for industrial data acquisition and PLC communication via OPC-UA, Modbus TCP/RTU, FINS Serial and FINS Ethernet protocols.
- Develop Angular frontend applications for real-time process monitoring, operator controls, recipe management, machine configuration and production reporting.
- Implement 21 CFR Part 11 compliance features — electronic audit trails, digital signature workflows and data integrity controls for regulated pharma environments.
- Package production Python applications as standalone Windows executables (PyInstaller) and register them as persistent Windows services using NSSM for zero-dependency deployment on production-floor machines.
- Lead junior developers — code reviews, coding standards, task assignment and technical mentoring across concurrent projects.

08/2025 – 12/2025

Junior Full Stack Developer

Quarks Systems · Ahmedabad, India

- Built and maintained RESTful APIs in Node.js/Express.js; integrated backend services with camera and printing hardware modules.
- Maintained Postman API documentation, improving cross-team integration efficiency by ~30%.

07/2023 – 12/2024

Full Stack Developer Intern — MEAN Stack

Fortray Global Services Limited · London, United Kingdom

- Built backend services in Node.js/Express.js with MongoDB, deployed to AWS EC2, with JWT authentication and RBAC.
- Worked in Agile sprints delivering end-to-end frontend/backend integration across multiple release cycles.

EDUCATION

01/2022 – 01/2023 **MSc Computer Science** — Greenwich University, London, United Kingdom

12/2015 – 01/2018 **Masters in Computer Applications** — Gujarat Technological University, Ahmedabad, India

12/2012 – 04/2015 **Bachelor of Computer Applications** — Gujarat University, Ahmedabad, India

KEY PROJECTS

L2 LineSync — WebSCADA System

Angular · FastAPI · Python · MySQL · PyInstaller · NSSM · 21 CFR Part 11

- L2 industrial process-monitoring platform — acquires real-time process data from the L1 server and displays live machine status, alarms and process values across production lines.
- Recipe management: full CRUD for recipes, per-machine recipe assignment, and per-machine / per-recipe production reports.
- User management: group-level and user-level CRUD, multi-tier RBAC (operator, supervisor, admin), configurable security policy, change-password functionality and feature-level access-rights control.

- Machine registration and configuration: register machines, configure parameters, manage machine-level settings; settings module includes database backup and restore for administrators.
- Digital signature functionality — users digitally sign off critical actions (recipe approvals, batch releases) with credentials that are cryptographically recorded, fully meeting 21 CFR Part 11 electronic signature requirements.
- L2 LineSync audit module: tamper-evident log of all system and user actions, with electronic audit trails and data integrity controls for pharma and regulated manufacturing compliance.
- Packaged the FastAPI backend as a standalone Windows executable using PyInstaller and registered it as a persistent background service via NSSM — enabling single-click deployment on production-floor machines with no Python environment dependency.

OPC Server — Custom Industrial Data Gateway

Python · Django · MySQL · OPC-UA · Modbus TCP/RTU · FINS Serial · FINS Ethernet

- Engineered a fully custom Python OPC server from scratch as a direct, open-source alternative to commercial OPC servers such as Kepware — delivering equivalent PLC connectivity and data acquisition capabilities at zero licensing cost.
- Implemented a pluggable, protocol-agnostic driver architecture with four production drivers: Modbus TCP/RTU, FINS Serial, FINS Ethernet and OPC-UA — allowing new device types to be onboarded by adding a driver module without touching core server logic.
- Reads live process data from PLCs through the driver layer and persists it into MySQL with a normalised schema supporting historical analysis, trend reporting and alarm logging.

Pulse Connect — OPC-UA Tag Browser & Config Generator

Python · OPC-UA (asyncua) · Angular · FastAPI · MySQL

- Full-stack industrial tool: connects to an OPC-UA server, browses the tag namespace, and lets engineers classify discovered tags as Process Tags or Alarm Tags through a guided UI — replacing a previously manual, error-prone process.
- Generates a structured config.ini file from the selected tag configuration, consumed directly by machine data-logging services with no further manual editing.
- Reduced commissioning time for new machines by enabling non-specialist engineers to configure data-logging end-to-end without editing raw configuration files.

Pharma Processing Machine — Configuration Tool

Python · config.ini · Industrial Protocols · Pharma Manufacturing

- Python utility that generates config.ini files for pharma processing machines — specifying IP addresses, database connection settings and trigger bits required for machine data-logging services to initialise correctly.
- Eliminates manual INI file editing errors through a structured, validated configuration workflow; deployed in a regulated pharma manufacturing environment ahead of production runs.

Compression Machine Interface — Angular SCADA

Angular · TypeScript · LabVIEW Connectivity · Industrial SCADA Software

- Angular-based SCADA application providing real-time operator screens for compression machines — machine status, process values, operator controls and parameter configuration.
- Defined the data interface contract between the Angular frontend and the LabVIEW backend layer — direct LabVIEW connectivity for faster real-time data exchange, with reliable state synchronisation and clean separation of concerns.
- Leading a junior developer on this project — component architecture, code review, integration testing approach and delivery planning.

Job Application — Job Search & Application Tracker

Python · FastAPI · PostgreSQL · React · TypeScript · Playwright

- Web app that collects software jobs in Germany from 75+ company career sites and recruiter job pages, removes duplicates and hides closed jobs.
- Scores each job against the CV, shows missing skills, and tracks applications, interviews and follow-ups.
- Creates a German/EU-format CV and cover letters, and fills company application forms automatically (Playwright).

TECHNICAL SKILLS

Programming	Python, TypeScript, JavaScript, FastAPI, Django, Node.js, Express.js, Angular, React
Industrial & OT	SCADA, WebSCADA, OPC, OPC UA, Modbus, FINS, PLC Communication, IIoT, 21 CFR Part 11, LabVIEW Connectivity
Databases & cloud	MySQL, PostgreSQL, MongoDB, AWS
Tools & practices	REST APIs, JWT, RBAC, PyInstaller, Windows Services, Docker, Git, Postman, Agile